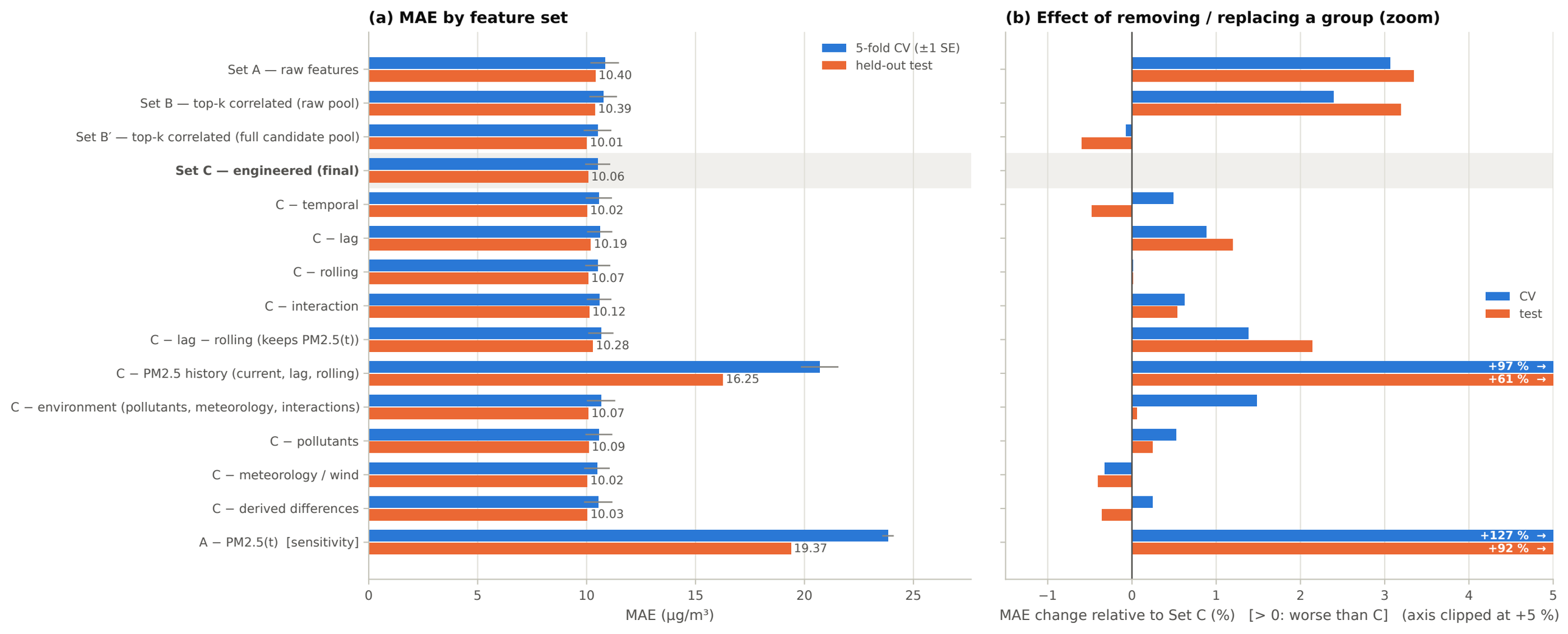
(a) MAE by feature set
5-fold CV (±1 SE)
held-out test
Set A — raw features
Set B — top-k correlated (raw pool)
Set B′ — top-k correlated (full candidate pool)
Set C — engineered (final)
C − temporal
C − lag
C − rolling
C − interaction
C − lag − rolling (keeps PM2.5(t))
C − PM2.5 history (current, lag, rolling)
C − environment (pollutants, meteorology, interactions)
C − pollutants
C − meteorology / wind
C − derived differences
A − PM2.5(t) [sensitivity]
10.40
10.39
10.01
10.06
10.02
10.19
10.07
10.12
10.28
16.25
10.07
10.09
10.02
10.03
19.37
0
5
10
15
20
25
MAE (µg/m³)
(b) Effect of removing / replacing a group (zoom)
CV
test
+97 % →
+61 % →
+127 % →
+92 % →
−1
0
1
2
3
4
5
MAE change relative to Set C (%) [> 0: worse than C] (axis clipped at +5 %)